Recent-Update GAN Experiment

Controlled HYB-MS comparison • identical real data • three paired seeds

MAIN RESULT

+10% GAN improves mean F1, but the advantage is not seed-consistent.

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
Updated real-only	93.08% ±0.56%	99.96% ±0.01%	67.35% ±2.67%	80.46% ±1.92%	99.48% ±0.07%	98.92% ±0.12%	0.01% ±0.00%	32.65% ±2.67%
Updated +10% GAN	93.63% ±0.46%	99.97% ±0.00%	69.96% ±2.19%	82.31% ±1.53%	99.48% ±0.10%	98.89% ±0.12%	0.01% ±0.00%	30.04% ±2.19%

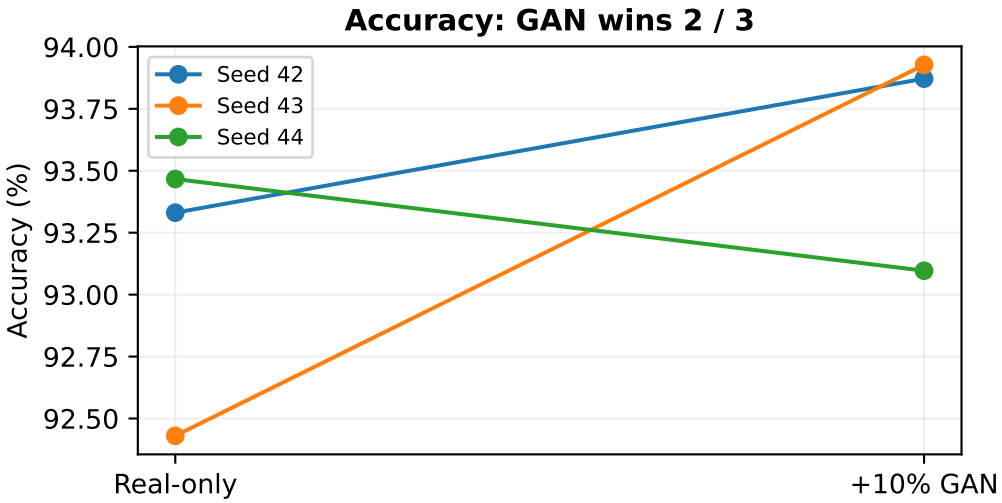
Mean GAN delta: F1 +1.85 points • recall +2.61 • FPR -0.0026 • ROC-AUC -0.003.

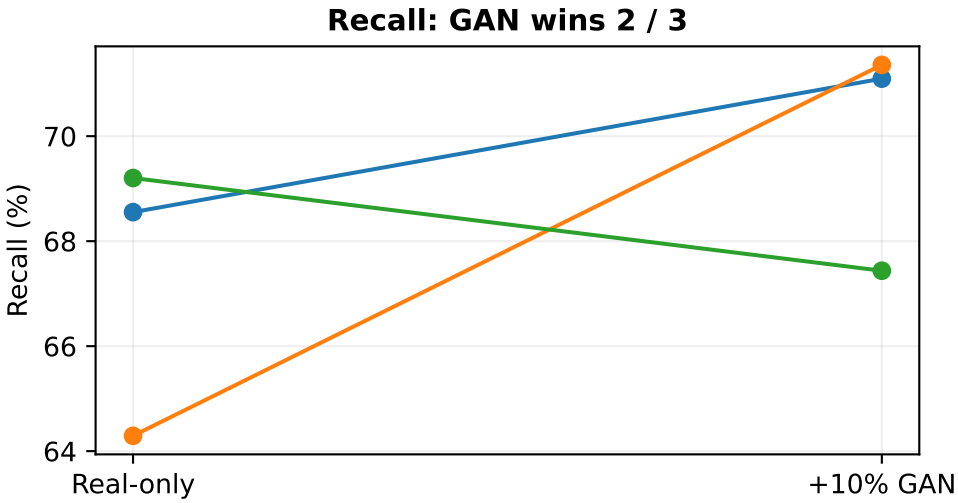
Interpretation: GAN augmentation helps the selected operating point on average, especially recall/F1, but seed 44 reverses that benefi
and mean ranking metrics do not improve. This supports a promising internal effect, not a robust deployment improvement.

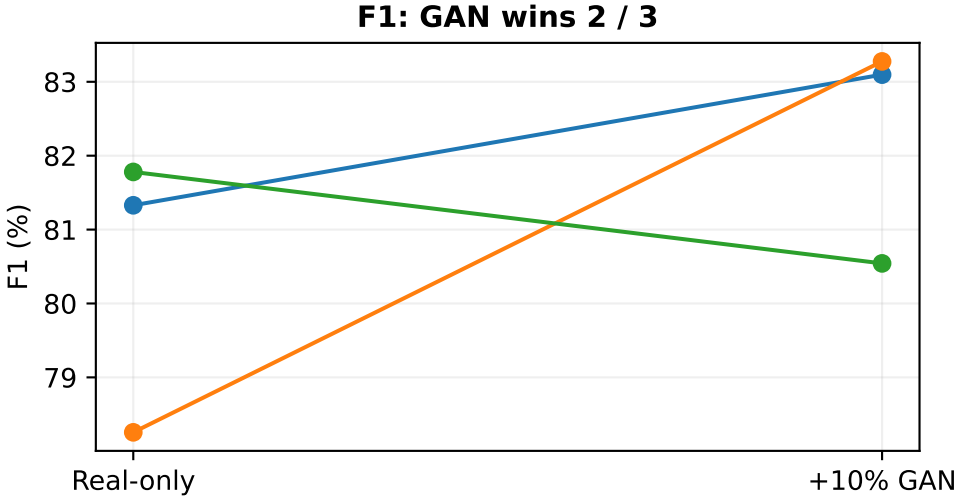
No protected confirmation data was used. Earlier external/calibration cohorts containing update identities are retired.

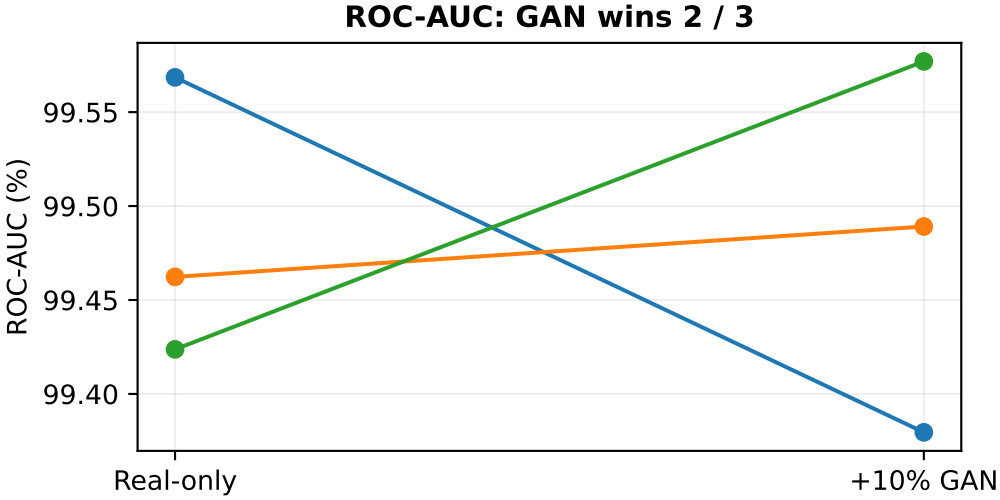
Paired Seed Evidence

Each line compares the same seed under real-only and +10% GAN









Method, Fairness, and Claim Boundary

What changed, what stayed fixed, and what this experiment can establish

FIXED ACROSS BOTH CONDITIONS

- 363,692 original multi-source real rows plus the same 3,492 recent-update rows.
- HYB-MS architecture, PSL-aware features, CORAL weight, weighted source sampling family, seeds 42/43/44, preprocessing, early stopping, metrics, and evaluation identities.
- 994 old validation rows and 602 old test rows were removed from both conditions because 42 and 35 registered domains, respectively, entered the update.

ONLY EXPERIMENTAL DIFFERENCE

- The GAN condition adds the already frozen 40,410 MS-HardGAN-C URLs and targets 10% synthetic exposure; the real-only condition adds none.

SUPPORTED CONCLUSION

- On the filtered internal test, +10% GAN raises mean F1 from 80.46% to 82.31% and mean recall from 67.35% to 69.96%, while lowering mean FNR from 32.65% to 30.04%.
- The F1/recall improvement occurs in two of three paired seeds, so it is promising but not stable.

NOT SUPPORTED

- The experiment does not show deployment readiness or temporal generalization.
- It does not prove GAN improves ranking quality: mean ROC-AUC and PR-AUC are slightly lower.
- The failed confirmation set cannot be rerun; a newly collected calibration and later confirmation are required.